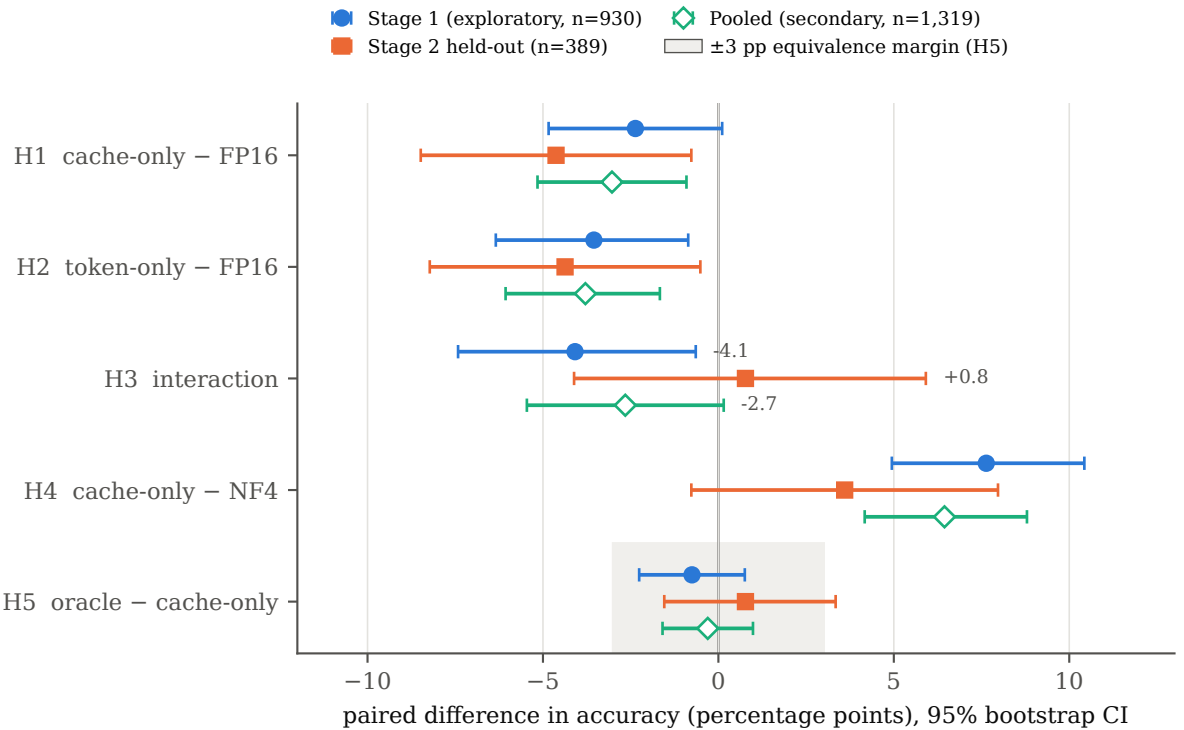
Stage 1 (exploratory, n=930)
Stage 2 held-out (n=389)
Pooled (secondary, n=1,319)
±3 pp equivalence margin (H5)
H1 cache-only − FP16
H2 token-only − FP16
H3 interaction
H4 cache-only − NF4
H5 oracle − cache-only
-4.1
+0.8
-2.7
−10
−5
0
5
10
paired difference in accuracy (percentage points), 95% bootstrap CI